

Date: 2026-06-15
Request ID: 100048653

Microwave				
Element(s)	Pt			
SOP#				
Acid Cocktail	9 mL HCl, 3 mL HNO3, 6 mL H2SO4			
Rack	7			
Vessel	Quartz			
Stir	90%			
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)
15	1200	180	65	110
15	1500	240	65	110
30	1500	240	65	110
sample(s)	weight (g)	volume (mL)		
200128299_1	0.5087	50		
200128299_2	0.5110	50		

Hotplate		
Element(s)	Pd	
SOP#	Determination of Palladium in Catalyst with or without Ag present	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128299_1	1.0032	100
200128299_2	1.0063	100

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128299	25.9790	27.9848	27.8965	4.40	0.9560

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm [dried]	%Pt [dried]
200128299_1	10/9	0.934	106.70	0.01
200128299_2	10/9	0.945	107.47	0.01

pt result: 107.09 ppm

pt result: 0.01 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128299_1	10/9	1.397	161.85	0.02
200128299_2	10/9	1.110	128.21	0.01

pd result: 145.03 ppm

pd result: 0.01 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$